

Joel Teply

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Profile

Twenty years building real-time perception systems that run on hardware which should not be able to run them. Semantic segmentation, monocular depth, surface normals, optical flow, camera pose and calibration, plane fitting, multi-frame fusion, scene reconstruction — all of it shipped on-device, in production, to millions of users, years before the platforms that later made it routine.

The recurring problem is the one an autonomous system lives with: perception components finish at wildly different times, and the world model cannot wait for the slowest. I build the representation, the scheduler, and the runtime around that constraint.

Experience

Cambrian Tech — Co-founder, CTO, Principal Engineer · 2009–present

Co-founded a computer vision and augmented reality technology company building proprietary perception technology and delivering custom engineering programs for enterprise clients across automotive, retail, and home improvement. Owned technical architecture and implementation end to end — ML, computer vision, real-time 3D perception, GPU optimization, native and web platforms — setting technical direction and working with engineering teams and customers to turn hard perception problems into deployable systems. Technology licensed to Fortune 500 customers including Home Depot and Lowe's, deployed to millions of users.

Automotive computer vision — Garmin, 2020–2022. Contributed to early perception research and engineering for an automotive program under strict accuracy and CPU/GPU constraints, applying geometry, pose estimation, sensor fusion, and optimization techniques developed for real-time spatial understanding. The program expanded into a multi-year engagement.

Real-time asynchronous perception. Maintained a continuously evolving scene representation at frame rate while expensive CNN inference ran asynchronously on-device at 1–3 second latency. Cellular-automaton region evolution over learned patch affinities, watershed segmentation across detected structural lines, tracking and temporal interpolation continuously added, removed, extended and joined regions; delayed semantic-segmentation and surface-normal results revised

the live state as they arrived rather than gating frame rate. Segmentation and surface normals both ran on the device — no cloud.

Two kinds of consistency, held at once. Geometric accumulation and tracking for spatial consistency — optical flow, camera pose, RANSAC plane fitting, multi-frame fusion, SLAM-like structure from an ordinary camera, camera parameter estimation (field of view, rotation), panorama-to-pinhole reprojection, online calibration in C++. Asynchronous state revision for perceptual consistency despite slow inference.

The runtime. Priority-scheduled, backpressured C++ perception pipeline over lazily evaluated shared frame representations, so estimators with radically different cost ran at their own cadence without stalling the world model. SIMD, threading, OpenGL ES compute and GPGPU shaders. One core architecture targeting iOS/Objective-C++, Android/JNI, Unity, browser, and headless servers.

Walkable AR flooring, before ARKit existed. A rendered floor that stays locked to the room as you walk on it — large, low-texture, steeply foreshortened surfaces where plane fitting fails and only real-time semantics hold the boundary. Required building the tracking and geometry ourselves.

Photographic compositing rather than overlay. Scene lighting and shadow estimation, texture and material decomposition, and re-rendering surfaces under the room's real illumination. The realism bar came from a visual-arts background: knowing what makes a composite read as a photograph rather than a sticker determined which physical quantities the perception stack had to recover.

Perception ML. Learned-affinity segmentation before AlexNet — a modified Siamese network learning the patch-similarity metric that drove region evolution; on our task it outperformed ImageNet-pretrained semantic segmentation and ran far faster. ICNet retargeted for both segmentation and surface-normal prediction, trained and exported for mobile inference. Soft segmentation and matting for boundary quality. Weight quantization and model compression for deployed CNNs (2017). Caffe→TensorFlow and Caffe→CoreML conversion pipelines. Synthetic training data from Unreal Engine scenes where real ground truth was unobtainable. Production Python/TensorFlow CV services for scene decomposition and reconstruction, with S3-backed data flow, Docker CPU/GPU deployment, testing/model utilities, and elastically scaling AWS inference. Inpainting and seamless texture synthesis.

First-named inventor, **US10964097B2** — *Pattern Recognition Systems and Methods for Performing Segmentation on Surfaces and Objects*.

EyeVerify / ZOLOZ — Software Engineer, Computer Vision & Biometrics · 2012–2016

One of the first two engineers on eye-vein biometric authentication; acquired by Ant Financial (Alibaba) for over \$100M in 2016.

- Scleral vasculature identification from commodity front-facing cameras, verification in under a second: eye and iris tracking, segmentation, multi-frame fusion, image enhancement at camera frame rate via OpenGL/GLSL and threaded pipelines.
- Presentation-attack and liveness detection. Co-inventor, **US9665784B2** — spoof detection using only a phone's earpiece and microphone (sonic 3D face sensing, photometric analysis, multi-source pulse detection). Also named on **US8437513B1**, **US8675925B2**.

TripleBlind → Ideem — Software Engineer, Cybersecurity · 2023–2026

Cryptographic authentication: device-bound key handling without dedicated secure hardware, secure client integration, and the SDK surface enterprise customers build against.

Earlier

Propaganda3 (2010–2012) · **VML** (2006–2010) · **AgileWise** (2005–2006, PCB design, touchscreen embedded prototype, RFID, Windows CE platform and display drivers) · **NovaStar**, **Federal Home Loan Bank of Topeka**, **AllofE Solutions** (1999–2006)

Current Work

Continuum — open-source Rust platform for AI agents running across commodity hardware. Sole architect, 550,000 lines. Inference runtime engineering (121 commits on a maintained llama.cpp fork: backend lifetime, KV-cache behaviour, Metal/CUDA placement), scheduling under memory scarcity, and embodied presence — fourteen concurrent 3D agents at 30 fps rendered by the same machine serving their models. github.com/CambrianTech/continuum

Sentinel-AI — transformer architecture research, *Experiential Plasticity*: attention heads that contribute specialize while the rest are pruned, so architecture co-evolves with training. 24% perplexity improvement on a 4B from a smaller model; models published with recipes. github.com/CambrianTech/sentinel-ai

Technical

Perception — semantic segmentation, monocular depth, surface normals, optical flow, camera pose and calibration, RANSAC plane fitting, multi-frame fusion, scene reconstruction, SLAM-like systems, tracking, matting, real-time compositing

ML — CNN architectures for segmentation, depth, normals, classification and reconstruction; metric learning and Siamese networks; GANs, diffusion and generative pipelines; synthetic data; transformers and LLMs; structured pruning, quantization, LoRA, inference-runtime engineering

Systems — C++, Rust, Python, Go, TypeScript; production Python ML services, Docker/AWS; SIMD, concurrency, GPU/GPGPU (Metal, CUDA, OpenGL/GLSL, WebGL), real-time scheduling and backpressure, memory-constrained execution, embedded and edge platforms

Platforms — iOS/Objective-C++, Android/JNI, Linux, Unity, Unreal Engine, browser ML, native camera pipelines, PCB-level hardware

Education & Patents

B.S. Computer Engineering — University of Kansas

- **US10964097B2** — *Pattern Recognition Systems and Methods for Performing Segmentation on Surfaces and Objects* (2021, Cambrian Tech LLC). First-named inventor.
- **US9665784B2** — *Systems and Methods for Spoof Detection and Liveness Analysis* (2017).
- **US8437513B1, US8675925B2** — *Spoof Detection for Biometric Authentication*.
- \$100,000 LaunchKC grand prize (2017); top-five finalist, Atlanta Startup Battle 6.0 (2019, TechSquare Labs).